Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 1 · SLO 2 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 1 : SLO â€“ 2 | Inserting an Element into an Array

I. Pseudocode â€” Append VAL

- Step 1: Set upper_bound = N + 1 (the array grows by one; new upper bound)
- Step 2: Set A[N + 1] = VAL (place the value at the new last position)
- Step 3: EXIT

II. Pseudocode â€” Insert VAL at position POS

- Step 1: [INITIALIZATION] SET I = N
- Step 2: Repeat Steps 3 and 4 while I >= POS
- Step 3: SET A[I + 1] = A[I] (shift element right)
- Step 4: SET I = I â^' 1
- [END OF LOOP]
- Step 5: SET N = N + 1
- Step 6: SET A[POS] = VAL
- Step 7: EXIT
- Note: shifting starts from the last element downward so no element is overwritten before being copied.